# 01 The Copying Room

## Mutation & accumulated change

| CORE PATH · BSC | EXTENSION · MSC | BEFORE THIS ROOM |
|---|---|---|
| **about 30 min** | **+ 28 min** | Nothing. This is the first room, and the place to start. |
| steps 1–4 | steps 5–7 | |

**OPEN THE ROOM**

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#copying`

One figure is copied, the copy is copied, and so on. Every copy carries a small random error, and nothing in the room is aiming at anything. This guide asks what accumulates, how fast, and in which direction.

### LEARNING OBJECTIVES

*After the core path you should be able to*

- Describe mutation as a copying error that is random with respect to any goal.
- Explain why lineages that start identical end up different from one another, with no selection involved.
- Predict how the size of each copying error changes the rate of divergence.
- Explain why a cloud of 50 lineages spreads out without its average moving anywhere.

*The extension adds*

- Use $\Delta \approx \sigma\sqrt{t}$ to predict divergence from the origin, and the $\sqrt{2}$ factor between pairs of lineages.
- Explain why bounded traits make divergence saturate, and what that does to divergence as a clock.

### SUGGESTED ROUTES

- **30-minute practical (BSc):** steps 1–4, then questions 1–2.
- **Lecture warm-up (10 minutes):** step 2 on a projector, the class predicting before each run.
- **Self-study:** core path and the *Checking your reasoning* page, about 35 minutes.
- **MSc:** steps 5–7 and questions 4–6, best paired with the Branching Room guide.

**How to work through it.** Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

## 1 One lineage, copied a hundred times

6 min

**SET UP**
The defaults: Polygon, 1 lineage, Mutation size 3, Stop after 100 copies.

**PREDICT FIRST**
After 100 copies, will the shape have changed a lot, a little, or not at all? Will the changes have pushed it consistently in one direction (bigger, rounder, more pointed…)?

**RUN**
Press **Start copying**. When it finishes, drag the History Scrubber slowly from 0 to 100.

**LOOK AT**
the card's *Δ from origin*, and how the shape changes generation by generation.

**EXPLAIN**
Was there a direction to the change? What in the model could have given it one, and does the model contain any such thing?

## 2 Three lineages from the same start

8 min

**SET UP**
Lineages in parallel: **3**. Everything else as before.

**PREDICT FIRST**
All three start from the identical figure and follow identical rules. After 100 copies, will they still look alike?

**RUN**
Press **Start copying**. Then press **New starting shape** and run again, and once more with the Fish.

**LOOK AT**
the three cards side by side, and the *pairwise divergence* line under the stage.

**EXPLAIN**
The three lineages had the same start and the same rules. What is the only thing that made them different?

## 3 Small errors, large errors

8 min

SET UP

3 lineages. Mutation size **1**, then Mutation size **5**.

PREDICT FIRST

Write down roughly how different the three lineages will be after 100 copies at each setting. Will size 5 give five times the divergence of size 1?

RUN

Run at size 1, then at size 5. Use the History Scrubber to compare both runs at generation 25.

LOOK AT

Δ from origin on each card, and the pairwise divergence.

EXPLAIN

Larger errors produce faster divergence — but did divergence keep climbing steadily, or did it start to level off at size 5? Note what you saw; step 7 explains it.

## 4 The whole cloud of possibilities

8 min

SET UP

Lineages in parallel: **50**. Mutation size 3.

PREDICT FIRST

Fifty lineages from one starting shape. After 100 copies, will the *average* shape have moved away from the start? Will the cloud of shapes be wider, narrower, or the same width as at generation 10?

RUN

Press **Start copying** and watch the cloud form. Scrub back to generation 10, then forward to 100.

EXPLAIN

Why does the spread of the cloud grow while its centre stays put? Relate your answer to the room's statement that mutation is unbiased.

## 5 The square-root law

10 min

**SET UP**
1 lineage, Mutation size 3 ($\sigma \approx 0.041$ of each trait's range), Stop after 100.

**PREDICT FIRST**
If every trait takes an independent random step of size $\sigma$ each generation, $\Delta$ from origin should grow as $\sigma\sqrt{t}$. Calculate $\Delta$ at $t = 25$ and $t = 100$. How many generations should it take to double the $\Delta$ you have at $t = 25$?

**RUN**
Run five times (press **New starting shape** between runs). For each, scrub to generation 25 and 100 and write down $\Delta$ from origin.

**LOOK AT**
the average of your five values at each time, against your prediction.

**EXPLAIN**
Why does doubling the difference take four times as long? Did the fit hold as well at 100 as at 25?

## 6 Two lineages diverge faster than one drifts from its origin

10 min

**SET UP**
3 lineages, Mutation size 3, Stop after 50.

**PREDICT FIRST**
The pairwise divergence line compares lineages with each other rather than with the origin. Should it read the same as the cards' $\Delta$ from origin, or larger? By what factor?

**RUN**
Run three times and record both numbers at the end of each run.

**EXPLAIN**
Account for the factor you found, using the idea of two independent random walks.

## 7 The ceiling

8 min

**SET UP**

3 lineages, Mutation size **5**, Stop after **400** copies.

**PREDICT FIRST**

If you run four times longer, will the lineages end up four times as different, twice as different, or something else?

**RUN**

Run, then scrub through to watch Δ over time.

**EXPLAIN**

The room clips each trait at the edge of its drawable range. What does that do to Δ late in a run, and why would it make two very old lineages look no older than two moderately old ones? (The Branching Room runs into exactly this.)

### DISCUSSION AND WORKSHEET QUESTIONS

1. BSc "Evolution is progress towards better forms." What does this room say about that claim, and what can it *not* say?
2. BSc Two lineages that began identical are different after 100 copies. List every cause of the difference that this model contains.
3. BSc The room says mutations "are not bad or good *per se*". What would have to be added to the model before *good* and *bad* meant anything?
4. MSc Divergence grows as $\sqrt{t}$, not as $t$. What does that imply for anyone trying to date a split from how different two forms look?
5. MSc In a real organism, traits are linked by development (a longer body usually means longer fins). How would correlated mutations change the cloud of 50 lineages?
6. MSc Why does clipping at a boundary, rather than reflecting from it, still let the cloud spread evenly at first but not later?

## CHECKING YOUR REASONING

*Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.*

- **Step 1.** The shape changes visibly but wanders: some features grow, others shrink, and the direction keeps changing. Nothing in the model favours any direction, because there is no selection and the mutation is unbiased.
- **Step 2.** The three lineages end up different from one another. The only cause is that each copying error is drawn independently in each lineage.
- **Step 3.** Size 5 diverges much faster than size 1 at first, but not five times as far by generation 100: the traits run into their bounds and Δ flattens.
- **Step 4.** The cloud widens steadily while its centre stays roughly where it started — the signature of unbiased change. The variance grows with time; the mean does not.
- **Step 5.** $\sigma\sqrt{25} \approx 0.21$ and $\sigma\sqrt{100} \approx 0.41$. Doubling Δ takes four times as long. At 25 generations the fit is good to a few per cent; by 100 the measured value runs about a quarter low because of the bounds.
- **Step 6.** Pairwise divergence should be about $\sqrt{2} \approx 1.41$ times Δ from origin: each lineage has walked independently, so the variances add.
- **Step 7.** Δ rises and then flattens a little under 0.5. Once traits sit at their bounds, more time adds no more difference, so a very old split cannot be told from a moderately old one.